

## Contact

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(LinkedIn)  
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## Top Skills

Artificial Intelligence (AI)  
Strategic Leadership  
Data Science

## Certifications

- 8. Practical Data Science Specialization (AWS | DeepLearning.AI) [3 Courses]
- 9. Apache Spark (TM) SQL with Databricks
- 7. Applied Data Science with Python Specialization (University of Michigan) [4 Courses]
- 6. AWS Fundamentals Specialization (4 Courses)
- 3. Cloud Digital Leader

## Honors-Awards

Dr. Wilber B. Payne Award for Excellence in Analysis  
Davis Rist Prize (Runner-Up)  
Chairman of the Joint Chiefs of Staff National Defense and Military Strategy Essay Competition (First Place)

## Publications

A Simulation-Optimization Approach to Estimate Workforce Requirements  
Big Data for Generals...and Everyone Else over 40  
A Markov Chain Model of Military Personnel Dynamics  
Optimizing Simulation Fidelity for Cost-Effective Aviation Training  
Artificial Intelligence: A Decisionmaking Technology

# Mark Zais, PhD, CAP-X

Enterprise Analytics Executive | Technical Leader | Data Science & Operations Research | AI Strategy & Implementation  
Greater Tampa Bay Area

## Summary

Enterprise analytics executive with 20+ years leading data strategy, AI innovation, and advanced analytics initiatives across defense, government, and industry. Trusted advisor to C-suite executives and senior government leaders on multi-billion-dollar investment, workforce, and capability decisions. I build and scale analytics organizations that transform data into strategic advantage. My approach combines deep technical expertise in operations research, machine learning, and decision science with proven ability to influence at the highest levels—from corporate boards to Congressional committees. What distinguishes my leadership is the ability to bridge executive strategy and technical execution: I set the vision, architect the solution, and work alongside data scientists and engineers to deliver results and translate complex quantitative insights into executive decision advantage.

**EDUCATION** ♦ Doctorate (PhD), Business Administration (Operations Research), University of Colorado at Boulder ♦ Master of Science (MS), Operations Research, Georgia Institute of Technology ♦ Master of Science (MS), Industrial Engineering, Georgia Institute of Technology ♦ Bachelor of Science (BS), Operations Research, United States Military Academy

**MORE ABOUT ME:** ♦ U.S. Army Veteran (Colonel) and former AH-64D Apache helicopter Pilot-in-Command. ♦ Certified Analytics Professional - Expert (CAP-X) ♦ Active TS-SCI security clearance ♦ Proficient in R, Python, SAS, SQL, LaTeX, MATLAB, and Power BI

**CORE CAPABILITIES:** Enterprise Analytics Strategy & Governance | Data Science & AI | Machine Learning & Predictive Analytics | Operations Research & Optimization | Decision Science | Executive Advisory & Stakeholder Influence | Cross-Functional Leadership | Business Development & Capture Management

**CAREER HIGHLIGHTS INCLUDE:** # Enterprise analytics strategy and AI governance for global 70,000+ person defense organization. # Led analysis team to optimize allocation of a \$13.5 billion annual budget across a \$40 billion budget horizon. # Analytics strategy for \$1.5B+ military modernization programs and

\$2B portfolio investments# Congressional and executive briefings influencing \$1B+ resource allocation decisions.# Board Member, INFORMS Analytics Certification Board.

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## Experience

Integration Innovation, Inc.

Senior Technology Director

August 2022 - Present (3 years 11 months)

Enterprise analytics executive responsible for data strategy, AI innovation, and advanced analytics capabilities across defense technology portfolios supporting a 900+ person workforce.

- ◆ Accountable for enterprise analytics strategy, operating model, and capability development across all business units and customer portfolios
- ◆ Architected analytics strategy for multi-billion dollar military modernization programs and portfolio investment decisions
- ◆ Drive business development and capture strategy, securing >\$600M in new contract awards through technical solution architecture and proposal leadership
- ◆ Established AI governance frameworks, analytics communities of practice, and delivery standards to institutionalize data-driven decision-making
- ◆ Trusted advisor to C-suite and senior government customers on digital transformation, AI strategy, and enterprise analytics priorities

## INFORMS

Board Member - Analytics Certification Board (ACB)

March 2025 - Present (1 year 4 months)

Oversee the Certified Analytics Professional (CAP) program as a standard-setting and assessment body for analytics professionals. The CAP certification provides independent verification of critical technical expertise and soft skills valued by analytics-oriented organizations globally.

## USSOCOM

Chief Data Scientist

June 2020 - June 2022 (2 years 1 month)

Tampa, FL

U.S. Special Operations Command. Enterprise AI and analytics strategy lead for a global command of 70,000+ personnel with a \$13B annual budget.

- ♦ Designed and led enterprise data science and machine learning capability supporting mission operations, workforce analytics, financial management, and medical research
- ♦ Established AI governance and analytics priorities aligned to enterprise digital transformation objectives
- ♦ Senior advisor to Command leadership on enterprise AI strategy, analytics investments, and data governance policy
- ♦ Built and mentored cross-functional teams of data engineers, data scientists, and ML engineers across mission domains

## Carnegie Mellon University

### Research Fellow, Information Technology and Strategy

July 2019 - June 2020 (1 year)

Collaborated with academia, industry, and government partners on AI policy, global strategy, and applied analytics research.

- ♦ First Place, 2020 Chairman of the Joint Chiefs of Staff National Defense Strategy Paper Award
- ♦ Nominated for three individual research writing awards focused on artificial intelligence strategy

## USSOCOM

### Strategic Studies & Analytics Branch Chief

August 2017 - July 2019 (2 years)

Greater Tampa Bay Area

Led enterprise modeling and analytics supporting strategic budget and acquisition decisions across a \$13.5B portfolio.

- ♦ Directed development of optimization platforms supporting \$40B in military construction planning
- ♦ Delivered executive-level analysis shaping long-range investment and resource allocation decisions

## Office of the Secretary of Defense

### Senior Operations Research Analyst

August 2014 - August 2017 (3 years 1 month)

Enterprise decision science advisor supporting defense-wide economic and budget policy within the Economic and Manpower Analysis Division (EMAD), Cost Assessment and Program Evaluation (CAPE) directorate.

- ♦ Led Congressionally directed studies and economic analyses influencing over \$1B in resource realignment
- ♦ Directed national-level compensation and housing allowance analytics optimizing billions annually
- ♦ Delivered findings to senior DoD leadership and U.S. Congressional committees

## University of Colorado at Boulder

### Doctoral Student

August 2011 - July 2014 (3 years)

- ♦ Earned a PhD in Business Administration (Operations Research).
- ♦ Research focused on the use of Simulation-Optimization, Dynamic Programming, and Metaheuristics (Graph Coloring, Tabu Search, GRASP) to solve complex real-world problems. [Java, SAS, SQL, MATLAB, SPSS]

## US Army

### Behavioral Forecaster | Operations Research Analyst

July 2008 - July 2011 (3 years 1 month)

Washington DC-Baltimore Area

U.S. Army, Deputy Chief of Staff G-1, Strength Analysis and Forecasting Division

- ♦ Led enterprise workforce forecasting and optimization for a 547,000+ employee organization, developing models to inform accession, retention, and recruiting policy during wartime operations.
- ♦ Presented Army-wide analytics directly to senior DoD leadership influencing critical wartime manpower policy decisions

## Georgia Institute of Technology

### Graduate Student

May 2006 - May 2008 (2 years 1 month)

Greater Atlanta Area

Earned two Master's Degrees: M.S. in Industrial Engineering & M.S. in Operations Research.

## US Army

### Aviation Officer / AH-64D Apache Longbow Pilot

June 1997 - May 2006 (9 years)

- ♦ Served in multiple leadership positions as an Aviation officer and helicopter pilot (Senior Aviator); designated as Pilot-in-Command (PIC).

- ♦ Commanded a company with 31 personnel and 8 AH-64D Longbow Apache helicopters and associated equipment valued at over \$250 million.

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## Education

University of Colorado Boulder - Leeds School of Business

Doctor of Philosophy (PhD), Business Administration (Operations Research)

Georgia Institute of Technology

M.S., Industrial Engineering

Georgia Institute of Technology

M.S., Operations Research

United States Military Academy at West Point

B.S., Operations Research